

ASPIRE COLLEGE VEHARI CAMPUS

Notes • BS CS 3rd Semester • Lecture 12: Switching Techniques (Circuit, Message & Packet Switching)

Course: Computer Networks (Credit Hours: 3)

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1. WHAT IS SWITCHING? (CONNECTING MULTIPLE NODES)

When thousands of computers want to communicate, connecting every computer directly to every other computer with dedicated cables is impossible. Instead, networks use intermediate forwarding devices called **Switches** to dynamically build paths between senders and receivers.

2. THREE MAIN SWITCHING TECHNIQUES

1. Circuit Switching

A **dedicated physical path** is established before communication begins and remains reserved until ended.

- 3 phases: Setup → Data Transfer → Teardown.
- Guaranteed bandwidth, zero mid-call delay.
- **Downside:** Inefficient; line remains locked even when silent.
- **Example** Traditional Landline Telephone System (PSTN).

2. Message Switching

Transmits the entire data block as a **single complete message** from node to node.

- Uses **Store-and-Forward** mechanism.
- Each intermediate switch stores the full message on disk before forwarding.
- **Downside:** High storage requirements; huge delay. (Obsolete today).
- **Example** Legacy Telegraph & Early Postal Mail systems.

3. Packet Switching

Data is broken down into small, manageable chunks called **Packets**.

- Each packet carries source/destination addresses.
- Packets travel independently through different routes.
- High link utilization; packets reassembled at receiver.
- **Example** Modern Internet (IP Networks).

3. TWO APPROACHES OF PACKET SWITCHING

A. Datagram Approach (Connectionless - Real Internet / IP):

- Each packet is treated as an independent entity (a **Datagram**).
- Packets belonging to the same message can take completely different paths depending on network traffic, and may arrive out of order.

B. Virtual Circuit Approach (Connection-Oriented - ATM / Frame Relay):

- A logical, predefined path (Virtual Circuit) is planned out before data transmission starts.
- All packets follow the exact same path in sequential order, but the physical link remains shared with other users.

4. SUMMARY COMPARISON: CIRCUIT SWITCHING VS. PACKET SWITCHING

Feature	Circuit Switching	Packet Switching
Dedicated Path	Yes (exclusive physical path reserved)	No (shared dynamically among all users)
Bandwidth Efficiency	Low (idle lines cannot be shared)	High (dynamically fills available capacity)
Call Setup Delay	High initial setup time needed	Zero setup delay in datagram mode
Charging / Billing	Billed by time/duration (per minute)	Billed by data volume (MBs / GBs transferred)
Dominant Use	Analog voice telephone calls	Internet web browsing, emails, video streaming

5. QUICK STUDENT REVIEW QUESTIONS

Q1: Why is Packet Switching preferred over Circuit Switching for Internet data?

Ans: Because computer data is bursty (users read a page, pause, then click again). Circuit switching wastes line capacity during silent pauses, while Packet Switching lets other users transmit packets in those gaps.

Q2: Why do packets sometimes arrive out of order in Datagram Packet Switching?

Ans: Because routers inspect real-time congestion and may steer packet #2 across a faster route than packet #1. The receiving computer reassembles them using packet sequence numbers.